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AI Analysis:

1. GenAI Names: Gemini by Google and ChatGPT by OpenAI.
2. Both accessed via Google search.
3. Prompt:
   Grader note: From the original C++ code, names were removed for security reasons.
   I will provide C++ code, instructions for implementation, and finally the 5 puzzles included. We need to accomplish the following:
   Write the solutions for each puzzle file to a file named solutionN.txt, where N is the puzzle number.
   • Each solutionN.txt file should include the following:
   o If no solutions are found, a single line: “No solution found.”
   o If solutions are found, it should include all solutions followed by a single line: “Solutions found: n”, where n is the number of solutions found.
   o The solutions should be in the same format as the input files: puzzle1.txt – puzzle5.txt.
   This is a major change from the original since the original outputs to the command line now should be output to SolutionN.txt. I would also personally like to make the change to lower the number of irrelevant or overly complex lines under the justification of maintainability.
   Misc information: Compiled using g++
   Returned output should be in the form of block changes for analysis and weighing
4. Generated code included in respective .txts
5. I went about generating code very differently here; I specifically queried to make changes in the form of blocks of code so I can pick and choose changes. I chose ChatGPT, it accomplishes the goal by editing the existing I/O classes first, then working on maintainability and readability changes. Since neither changes the core logic of the Sudoku and both use around the same output method, they tie on time and space complexity. ChatGPT wins on complexity due to the simple blocks it outputs, as well as the large simplification it makes to the Sudoku app class; the rewrite removed a ton of unnecessary header and now deals exclusively with calling other functions. Gemini makes the pitfall of hardcoding the inputs and still iterating through them, while ChatGPT’s solution is more elegant, iterating through both at the same time. Google Gemini fails on correctness as well since it doesn't format output correctly. ChatGPT instead returns fully proper output.
6. I chose ChatGPT based on the dramatic difference in correctness and maintainability; the short, easy-to-understand blocks of output by ChatGPT allow for easy modification and easy implementation as well. Even to the point of allowing me to skip certain requested changes. In comparison, Gemini did full rewrites that made the code more complex and harder to read. So, based on correctness and maintainability, I chose ChatGPT.
7. Implementation successful. Changes made by hand include: ChatGPT did no rewrites for the print module, so I rewrote it by hand. Manually removed #include for vectors since

I removed that approach. Added multiple missing curly braces. Added extra error handling including couldnt read file.

8. Compiling and debugging were successful; some of the particulars are included above.